

Garv Arora

Computer Science Engineer | Intelligent Robotic Systems & Software

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Education

Vellore Institute of Technology (VIT) <i>Bachelor of Technology – Computer Science & Engineering (Specialization: IoT) CGPA: 8.36 / 10.0</i>	Vellore, India <i>Aug 2023 – Jul 2027</i>
Gems International School (CBSE) <i>Class XII: 76.4% Class X: 89.3%</i>	Gurugram, India <i>Nov 2016 – Apr 2023</i>

Work Experience

LG Soft India (LGSi) – HS/ES India Lab <i>Living Solution Control Development Intern</i> Constructed C/C++ embedded control logic and state machine handlers for living solution control modules, verifying execution timing on target hardware; work certified as “Technically Competent” by LGSi HR/Engineering management.	Noida, India <i>Jun 2026</i>
Samsung PRISM <i>Data Intelligence Agent Developer (Internship)</i> Engineered multi-agent LangGraph data engine (FastAPI, Redis, PostgreSQL, Qdrant) automating discovery and 20-stage analysis of dataset corpora; integrated spaCy NLP, Presidio PII, and MemorySaver HITL pause.	Remote <i>Apr 2026 – Oct 2026</i>
Wissen Baum Engineering Solutions LLP <i>Software Automation Intern</i> Authored Python BDD test automation framework (Gherkin/Behave, pytest, Playwright), parallelizing GitLab CI jobs to cut pipeline execution time from ~10m to ~6m (40% speedup) and reducing manual testing by 80%+.	Pune, India <i>May 2025 – Jul 2025</i>
SEDS India (SEDS VIT) <i>Autonomous Systems Developer</i> - Architected ROS2 Nav2 & RTAB-Map 3D Visual SLAM pipeline on Intel RealSense D455, generating real-time elevation maps and slashing manual teleoperation time by 70% in competition. - Slashed CV pipeline latency by ~60% and boosted throughput 4x using CUDA GPU kernels, YOLOv8, ArUco/AprilTag tracking, and Linux V4L2 zero-copy buffer tuning across 4 camera streams. - Flashed Micro-ROS firmware onto ESP32 microcontrollers for distributed sensor polling, fusing wheel encoders and 6-DOF IMU with an Extended Kalman Filter (EKF) to cut odometry drift by ~40% (sub-10 cm accuracy). - Shipped PyQt5 mission dashboard with real-time telemetry visualization, parameter tuning, and FSM-based autonomous mission execution.	Vellore, India <i>Apr 2024 – Jul 2026</i>

Projects

Captivity CLI <i>Linux D-Bus, systemd, Socket IPC</i> pypi.org/project/captivity-cli Built autonomous captive portal WiFi login daemon featuring <50ms HTTP 204 probing, systemd integration, D-Bus NetworkManager monitoring, keyring encryption, and dual Python-Rust TCP socket IPC.	2026
Gesture-Controlled 5-DOF Robotic Arm <i>C++, Arduino, MPU6050, Flex Sensors</i> Engineered a 5-DOF robotic arm driven by a sensor glove (MPU6050 IMU, flex sensors), mapping spatial gesture telemetry to multi-servo PWM angles with <50ms response latency and 94% gesture accuracy.	2025
3D Reconstruction – IMU-Enhanced KinectFusion <i>C++, OpenCV, Intel RealSense SDK</i> Stabilized KinectFusion TSDF reconstruction by fusing RealSense D455 depth and 6-DOF IMU data via complementary filter, reducing tracking failures by ~50% across fast-motion trajectories.	2025

Patents & Publications

Patent Published (App IN202641072249 A1): <i>Autonomous Radar-Guided Survivor Detection and Navigation System</i> — Fused 24GHz FMCW mmWave radar, ultrasonic array, and IMU dead-reckoning on dual-core ESP32; formulated confidence grid decay $C = \max(0, C - \beta \Delta t)$ for survivor breathing detection under rubble. (2026)

Achievements & Awards

Amazon ML Summer School 2026: Selected among Top 3,000 out of 1,30,000+ applicants across India (~2.3% acceptance rate).
International Rover Challenge (IRC): Ranked 13th Globally (2025) and 17th Globally (2026) with Team Vyadh (SEDS VIT).
International Rover Design Challenge (IRDC 2025): Top 5 Finalist with Team Vyadh (SEDS VIT).
TEDx Speaker: Delivered an official TEDx talk on challenging gender stereotyping and social equality.

Certifications

AWS Certifications: AWS Certified AI Practitioner AWS Certified Cloud Practitioner
Oracle Cloud Certifications: Certified Data Science Professional Certified Generative AI Professional

Technical Skills

Languages: C, C++, Python, Rust, ARM Assembly, Java, JavaScript, SQL, Bash, Verilog
Robotics & Autonomous Systems: ROS2 (Nav2, MoveIt2), RTAB-Map (3D SLAM), Micro-ROS, EKF/UKF, PID/MPC, Sensor Fusion, mmWave Radar
Perception & Machine Learning: OpenCV, YOLOv8, MediaPipe, Intel RealSense SDK, PyTorch, TensorFlow, scikit-learn, CUDA, Transformers
Embedded & Systems: ARM Cortex-M, ESP32, FreeRTOS, I2C/SPI/UART, Bare-Metal C/Assembly, Linux Kernel/D-Bus, Docker, Git, CI/CD, AWS